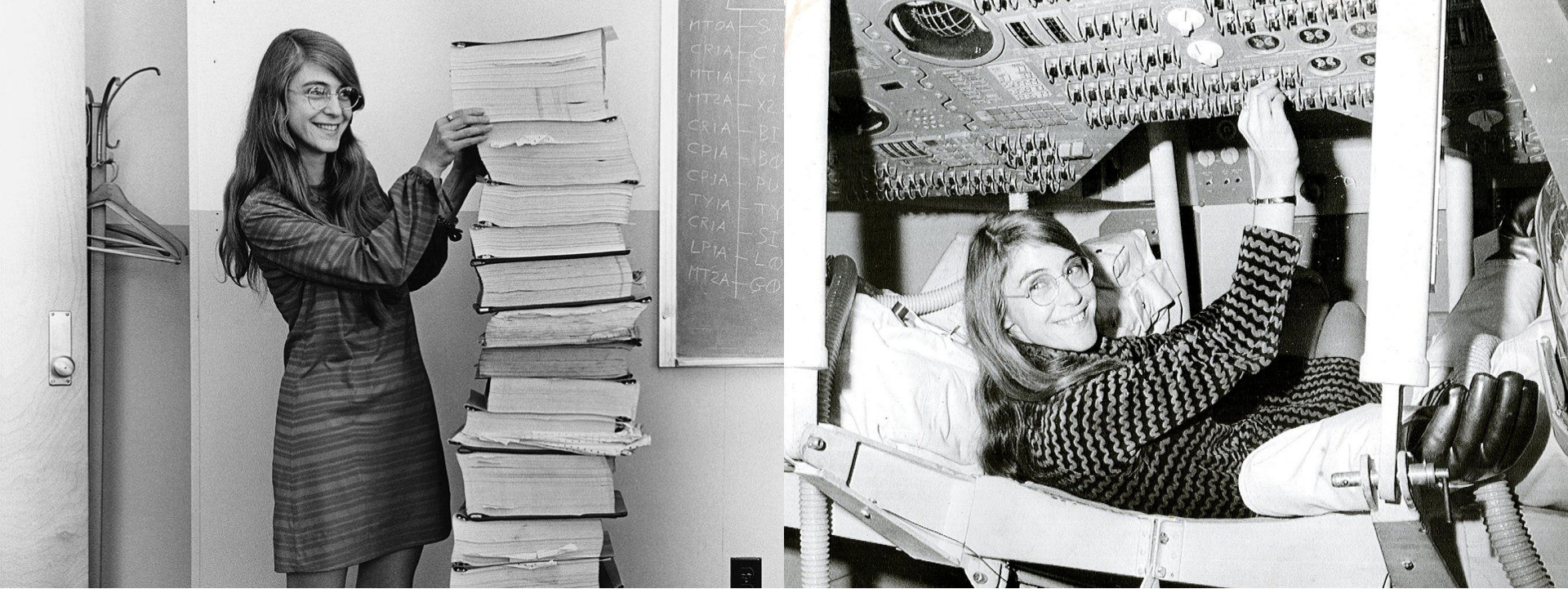


Software Engineering

Introduction

Who coined the term “Software Engineering”?



**Who coined the term
“Software Engineering”?**

Who coined the term “Software Engineering”?

Margaret Hamilton

- Born in 1936 in Indiana, USA
- Bachelor of Mathematics
- Self taught Computer Scientist
- Director of the Software Engineering Division of the MIT Instrumentation Laboratory
- Developed on-board flight software for NASA's Apollo program (1961 to 1972)
- Founded two software companies
- Published more than 130 papers, proceedings, and reports, about sixty projects, and six major programs
- Received the Presidential Medal of Freedom from US president Barack Obama for her work regarding NASA's Apollo Moon missions

[Source: Wikipedia](#)

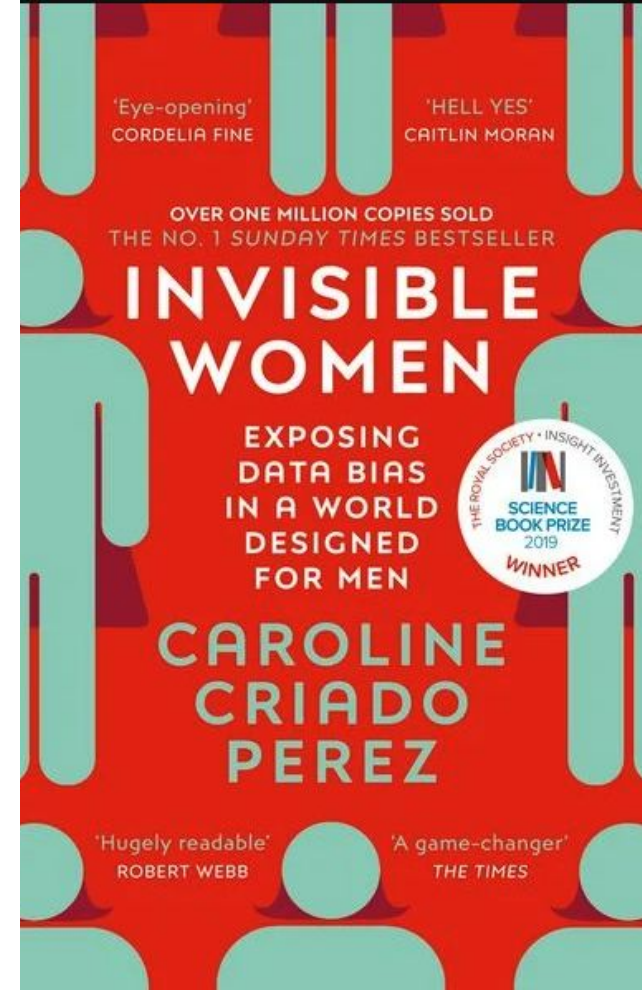


Further Reading

Invisible Women

Exposing data bias in a world designed for men

by Caroline Criado Perez

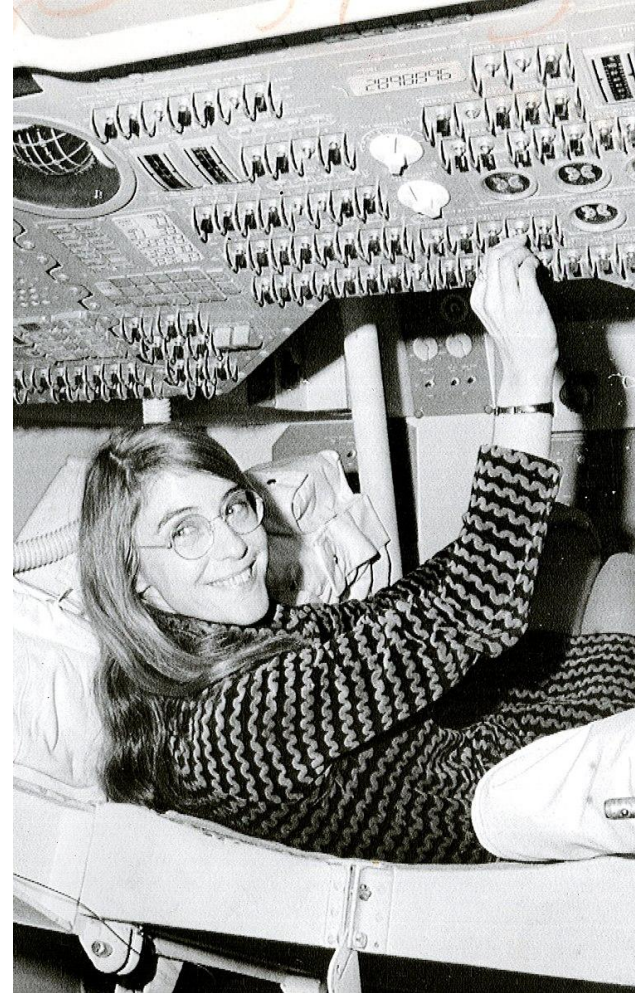


Who coined the term “Software Engineering”?

“I fought to bring the software legitimacy so that it—and those building it—would be given its due respect and thus I began to use the term ‘software engineering’ to distinguish it from hardware and other kinds of engineering, yet treat each type of engineering as part of the overall systems engineering process. When I first started using this phrase, it was considered to be quite amusing. It was an ongoing joke for a long time. They liked to kid me about my radical ideas. Software eventually and necessarily gained the same respect as any other discipline.”

– Margaret Hamilton

[Source: IEEE Computer Society](#)



Who coined the term “Software Engineering”?

“In the fall of 1968 and again the in fall of 1969, NATO hosted a conference devoted to the subject of software engineering. Although the term was not in general use at that time, its adoption for the titles of these conferences was deliberately provocative. As a result, the conferences played a major role in gaining general acceptance, perhaps even premature, for the term.”

[Source: Robert M. McClure, participant](#)

SOFTWARE ENGINEERING

Report on a conference sponsored by the
NATO SCIENCE COMMITTEE
Garmisch, Germany, 7th to 11th October 1968

Chairman: Professor Dr. F. L. Bauer
Co-chairmen: Professor L. Bolliet, Dr. H. J. Helms

Editors: Peter Naur and Brian Randell

January 1969

Software Engineering: Basic Terminology

Software Layers

Infrastructure

Database

Backend

Frontend

UI (User Interface)

Software Engineering: Basic Terminology

Hardware that Software is installed to

OnPremise Datacenters

Cloud Servers

Mobile Devices

Desktop PCs (Personal Computers)

What is “Software Engineering”?

Development Processes
Quality Management
Requirements Engineering
System Design
Architecture
Implementation
Testing
Delivery
Maintenance
Scaling
Operations

Exercise: 20 minutes

Software Engineering

- Development Processes
- Quality Management
- Requirements Engineering
- System Design
- Architecture
- Implementation
- Testing
- Delivery
- Maintenance
- Scaling
- Operations

... in the company you work(ed) for

Each of you draws a diagram explaining one of these aspects of Software Engineering in the company you work(ed) with.

Work in teams of 3 to 4 to discover similarities and differences.

